

ADARSH SAURABH

+91 7004428313 adarshsaurabh1396@gmail.com [LinkedIn](#) [GitHub](#) [Devpost](#) [LeetCode](#)

Education

National Institute Of Technology, Rourkela

Master of Technology in Signal and Image Processing, CGPA: 8.28

2025 – 2027

Rourkela, India

Guru Ghasidas University, Bilaspur

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.50

2020 – 2024

Bilaspur, India

Jawahar Navodaya Vidyalaya, JNV Sitamarhi

Matriculation CBSE, CGPA: 8.8

2017

Sitamarhi, India

Relevant Coursework

- Data Structures
- Algorithms Analysis
- Artificial Intelligence
- Computer Vision
- Software Methodology
- Database Management
- Computer Architecture
- Soft Computing

Experience

Autobot Robotics

May 2023 – Aug 2023

Machine Learning Intern

Remote

- Mentored and guided a junior intern, imparting mastery in ML fundamentals, Data Manipulation, NumPy, and Pandas.
- Developed and deployed (Django) 4 ML/DL projects with models for integration into company robotic systems.

Ziroh Labs [Source Link](#)

Jan 2022 – Feb 2022

Team Leader - Academic Alliance Program

Remote

- Fostered strong mentor-team connections, facilitating seamless communication and collaboration.
- Successfully led a cross-functional team of 5 to achieve image processing project completion ahead of schedule.

Key Projects

Amazon ML Challenge 2026 [Source Link](#) | Python, Polars, CatBoost GPU, LightGBM, RapidFuzz

2026

- Engineered end-to-end production entity resolution pipeline over 24.2M records using Polars lazy execution and multi-view blocking.
- Built 14-dim C++ RapidFuzz feature extraction engine; trained CatBoost GPU and LightGBM with Isotonic calibration achieving Macro F0.5 = 0.9719.

Alternative Data Radar [Source Link](#) | Python, Next.js, SQL, Bright Data, LLM APIs

2026

- Engineered automated data collection backend in Next.js to gather public web signals (careers and pricing data), routing requests via Bright Data Web Unlocker to bypass anti-scraping mechanisms.
- Implemented AI-driven analysis to calculate a 0–100 corporate health score stored in a SQL database, presenting pre-earnings intelligence and warning alerts in a responsive Recharts dashboard.

Apna Gold Solutions — Multi-Tenant SaaS Platform [Source Link](#) | Django REST API, React Vite, JWT

2025

- Developing multi-tenant B2B2C platform with seller registration, KYC verification, and customer subscription plans.
- Implemented custom JWT authentication with database conflict resolution and secure tenant data isolation.

Uplab — Adversarial Document Intelligence Pipeline [Source Link](#) | Python, Gemini API, LangGraph, Streamlit

2025

- Architected adversarial multi-agent workflow in LangGraph utilizing Gemini 2.5 Pro specialists to check visa document coherence; reduced manual verification workload by 85% with detailed failure rebuttal generation.

CodeAtlas — AST Architecture Analyzer [Source Link](#) | Python, AST, Graph Algorithms, Mermaid.js

2026

- Engineered zero-dependency static analysis engine using standard library `ast` to parse function hierarchies, imports, and cross-module dependencies without executing user code.
- Implemented automated anti-pattern detection for nested loops and $N + 1$ database queries, rendering interactive codebase architecture graphs via Mermaid.js in a local web workbench.

IBYD Technology — Warehouse PathMapper [Source Link](#) | Python, Algorithms, Data Analysis

2023

- Delivered dynamic warehouse mapping solution as a freelance engagement, working directly with client to translate operational pathfinding needs into a heuristic-based routing system.
- Scaled routing engine to warehouses as large as $10,000 \times 10,000$ units, computing optimal paths through 10,000+ locations simultaneously in under 0.5 seconds on a standard i5 laptop CPU.

K-HOG Unsupervised Keyframe Identifier (K-HUKI) [Source Link](#) | Python, Computer Vision, ML

2023

- Built keyframe extraction system with HOG Features and Unsupervised Learning achieving 96.45% accuracy.
- Developed preprocessing architecture 11 times faster than ResNet-based approaches with real-time processing.

Achievements

Intermediate Winner — Awaiken Hacks 2023 [Source Link](#): Innovative tool using MediaPipe for real-time action analysis in education with precise feedback on posture and motion.

Rank 1 — Girlsript Bilaspur Hackathon "Hack4CG" [Source Link](#): Won top honors for Facial Emotion Recognition project addressing depression vulnerability in modern society.

Leadership / Extracurricular

Google Developers Student Club

Vice President

3rd Year

Guru Ghasidas University

Technical Skills

Languages: Python, C, C++, HTML, CSS, SQL, TypeScript

Developer Tools: VS Code, Git, GitHub, CUDA, Anaconda

Technologies/Frameworks: Polars, CatBoost (GPU), LightGBM, RapidFuzz, Django, Flask, Next.js, React Vite, LangGraph, Streamlit, NumPy, Pandas, TensorFlow, Scikit-learn, OpenCV, Supabase, Docker

Specializations: Machine Learning, Deep Learning, Natural Language Processing, Large Language Models, LLMOps, Computer Vision, Multi-Agent Systems, Data Structures & Algorithms